WEEK 03 / AI ENGINEERING

AnomalyGuard — Transaction Anomaly Detection & Analysis

Monday 9 November – Friday 13 November 2026 | Five-day guided project

01 Real-world challenge

Financial and operational teams must review unusual transactions efficiently. Build a reproducible unsupervised-learning pipeline that highlights unusual records; an anomaly is a review signal, not proof of fraud.

02 Project objectives

- Load a manageable numeric transaction dataset and explore distributions.
- Clean and scale selected features without leaking evaluation information.
- Train an Isolation Forest and generate anomaly scores/flags.
- Visualize flagged records, explain model limits and compare threshold choices.

03 What you will learn

- Unsupervised anomaly detection and feature engineering
- Isolation Forest scores and threshold sensitivity
- Proper train/evaluation separation when labels exist
- Model limitations, false positives and reproducible notebooks

04 Recommended tools and flexibility

Python, pandas, scikit-learn, matplotlib/seaborn; Jupyter Notebook. Use public or synthetic transactions with clear provenance; no GPU or paid service.

Equivalent tools are welcome if the required learning outcomes and deliverables are met. Keep the project manageable within five working days.

05 Five-day implementation roadmap

Monday — Dataset and exploratory analysis

- Choose a documented public or clearly labeled synthetic transaction dataset.
- Inspect schema, missingness, distributions and plausible anomaly signals; define review objective.

Daily evidence: Dataset source, notebook and EDA charts.

Tuesday — Preprocessing and baseline

- Select features; handle missing values and categorical columns where needed.
- Scale appropriate features and define a simple statistical/rule-based baseline.

Daily evidence: Reproducible preprocessing cells and baseline findings.

Wednesday — Isolation Forest model

- Train Isolation Forest with documented parameters and random seed.
- Compute anomaly scores and rank suspicious records; inspect sample explanations based on feature values.

Daily evidence: Model outputs, top anomalies and interpretation notes.

Thursday — Thresholds and evaluation

- Compare at least two thresholds/contamination settings and chart resulting flag counts.
- If trustworthy labels exist, report suitable precision/recall; otherwise use case review and stability checks, not fabricated accuracy.

Daily evidence: Threshold comparison and limitations discussion.

Friday — Reproducibility and communication

- Clean notebook to run top-to-bottom and summarize practical findings and false-positive risks.
- Deliver source, requirements, dataset instructions, charts and narrated demo.

Daily evidence: Reproducible notebook, results summary and video.

06 Completion criteria

- Exploration, preprocessing, baseline and Isolation Forest implemented.
- Ranked anomalies and threshold comparison are interpretable and reproducible.
- Clearly state that anomaly scores do not establish fraud.

07 Optional stretch challenges (not mandatory)

- Compare Local Outlier Factor with care about prediction semantics
- Add a small Streamlit inspection interface
- Evaluate stability across random seeds

08 Daily submission and professional conduct

- Submit each day’s progress using the official Google Form by 11:59 PM PKT, even if unfinished; explain blockers honestly.
- Provide meaningful screenshots, source commits or editable design evidence and a short progress note.
- Use your assigned domain group for questions and mentor guidance; do not share passwords, tokens, personal data or private documents.

09 Beginner starter pack

- Begin with a small, well-described dataset.
- Keep a fixed random seed and record feature choices.
- Do not use final fraud labels as model input.

10 Friday handover checklist

- Provide a working GitHub repository for coding projects, or organized editable design source files for Graphic Design.

- Include a clear README or design case study: objective, tools, setup/opening instructions, features, limitations and testing.
- Attach final screenshots/exports and a concise list of known issues or unfinished optional features.
- Publish a professional LinkedIn project update, tag Stack & Scale, and follow the internship communication guidelines.

11 Mandatory Friday narrated video demonstration

- Show the actual project or clickable design in use, not only slides or static screenshots.
- Explain at least one important technical/design decision in your own words.
- Describe one real challenge, how you addressed it and what you learned.
- Keep the recording clear, accessible to mentors and consistent with the delivered source.

12 Responsible AI and original work

- AI assistance is allowed for learning, research and debugging, but you must understand and verify all submitted work.
- Do not submit generated output you cannot explain; acknowledge meaningful assistance and respect licenses.
- Test claims, models, API results and designs yourself; never fabricate data, results or user feedback.